

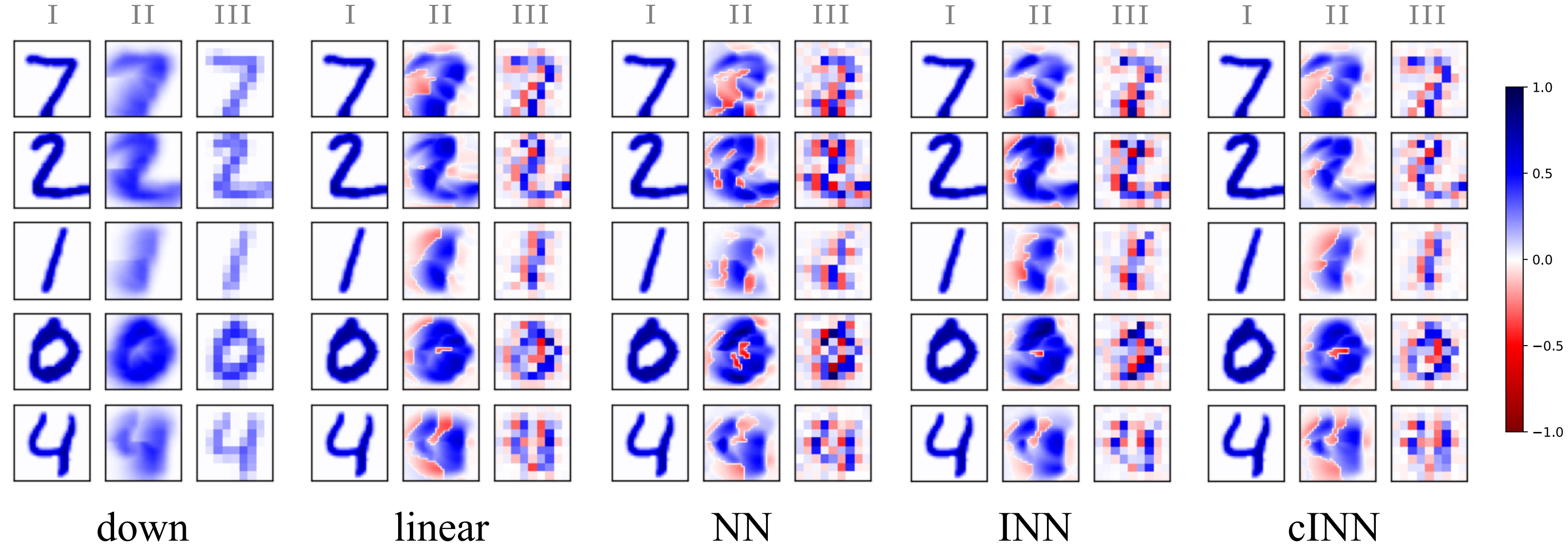
Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY

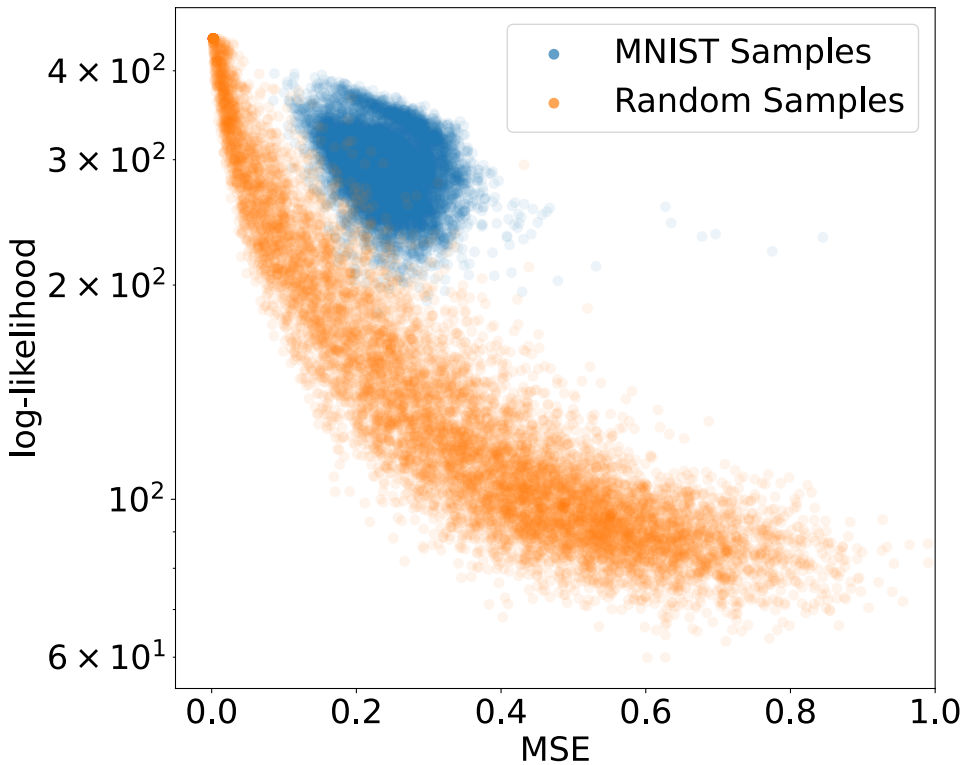
2

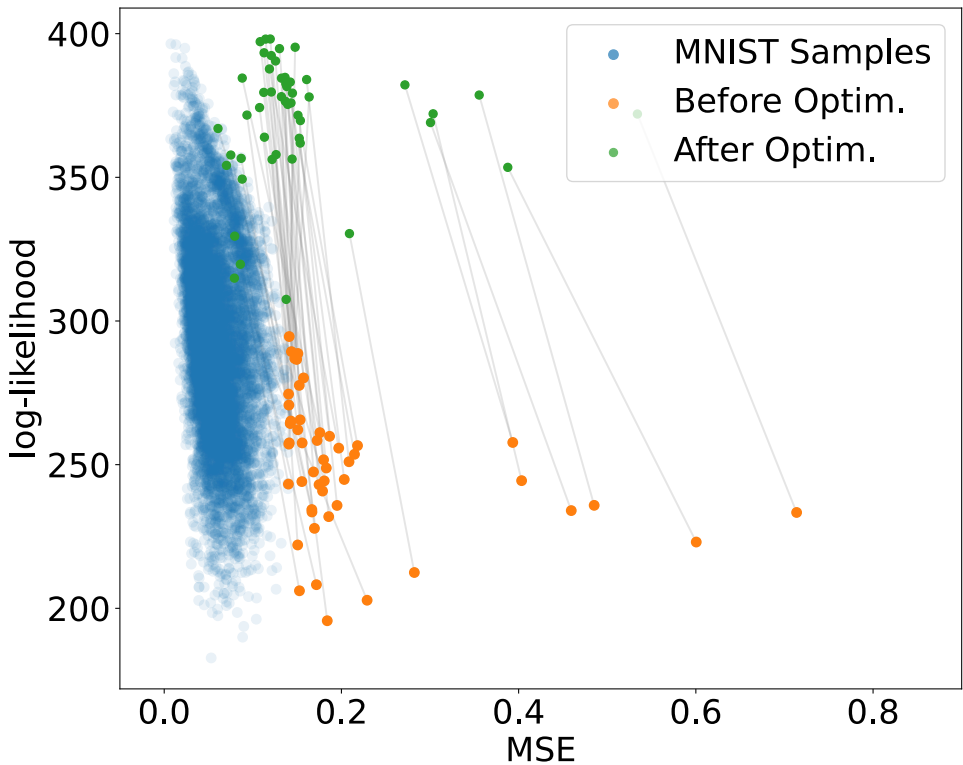
8

invertible Neural Networks



Methods	Res.	MAE ↓	SSIM ↑	PSNR ↑	ACC ↑	#Param ↓
down		0.3432	0.3784	25.0973	0.2884	1
linear	28	0.3279	0.4770	25.1453	0.5884	63,504
NN	×	0.3340	0.4901	25.2400	0.6192	4,270,900
INN	28	0.3231	0.4937	25.2729	0.5764	18,063,390
cINN		0.3136	0.4943	25.5161	0.5932	2,449,197

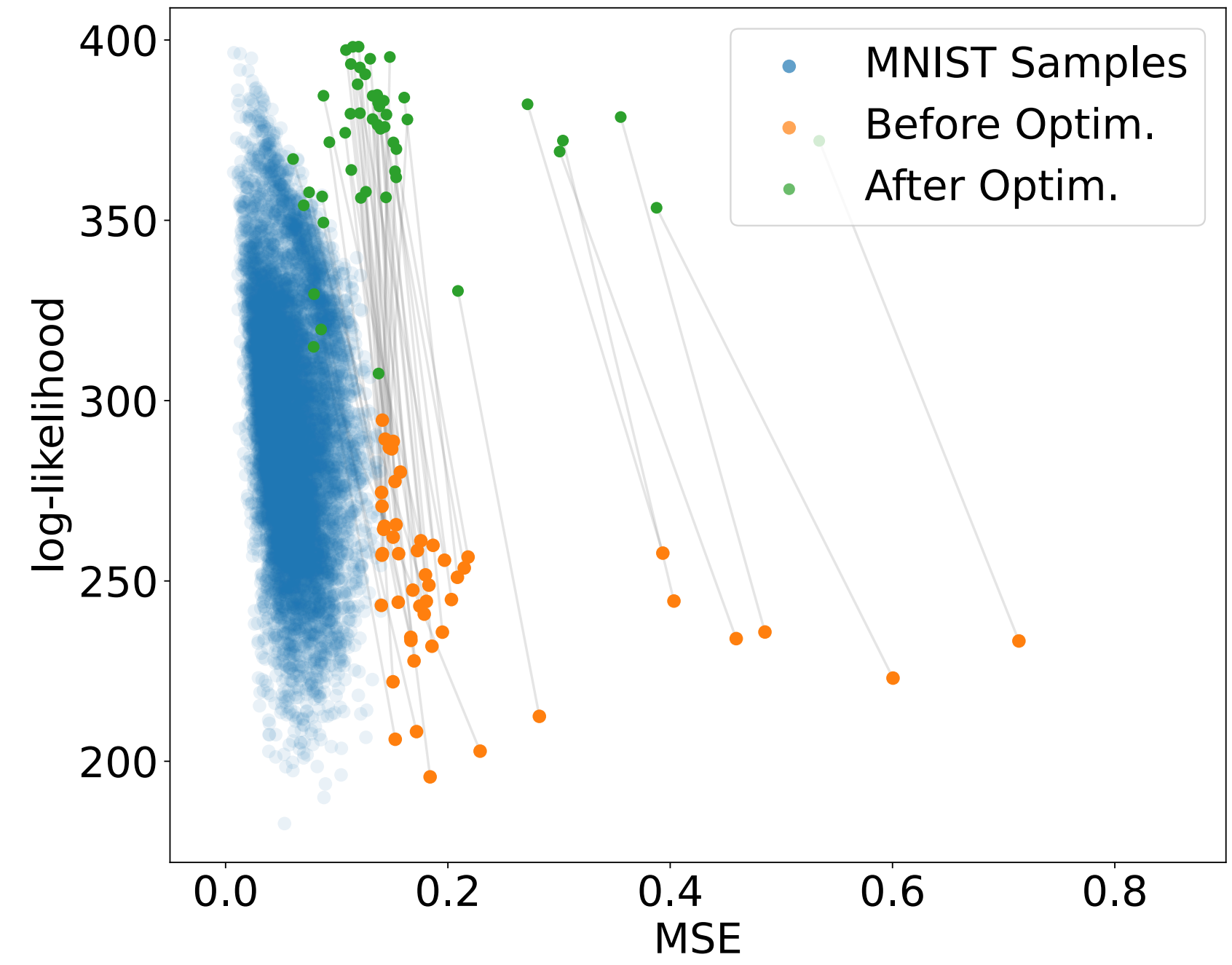
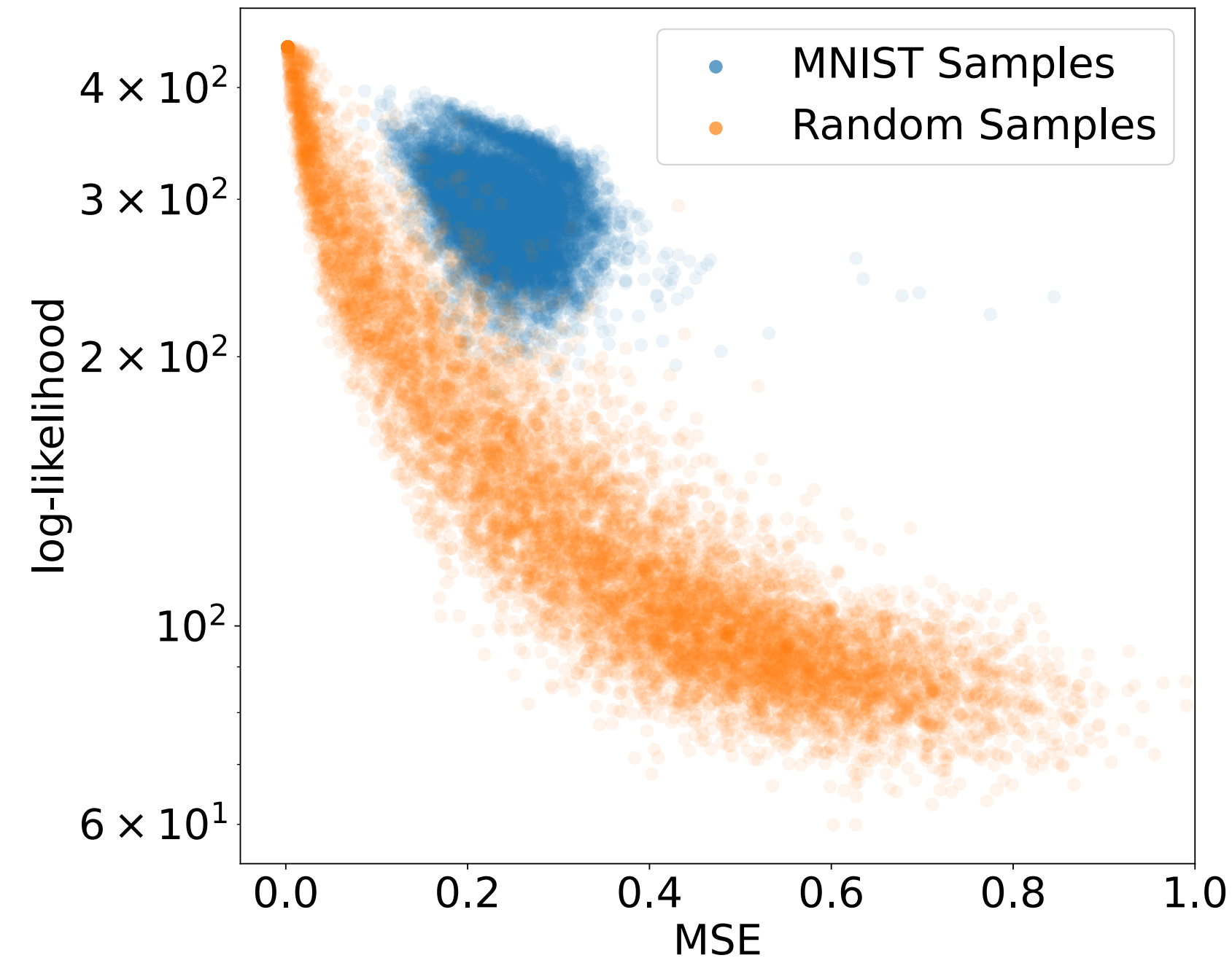




[Wuetai., 2024]

Additional alternative restriction

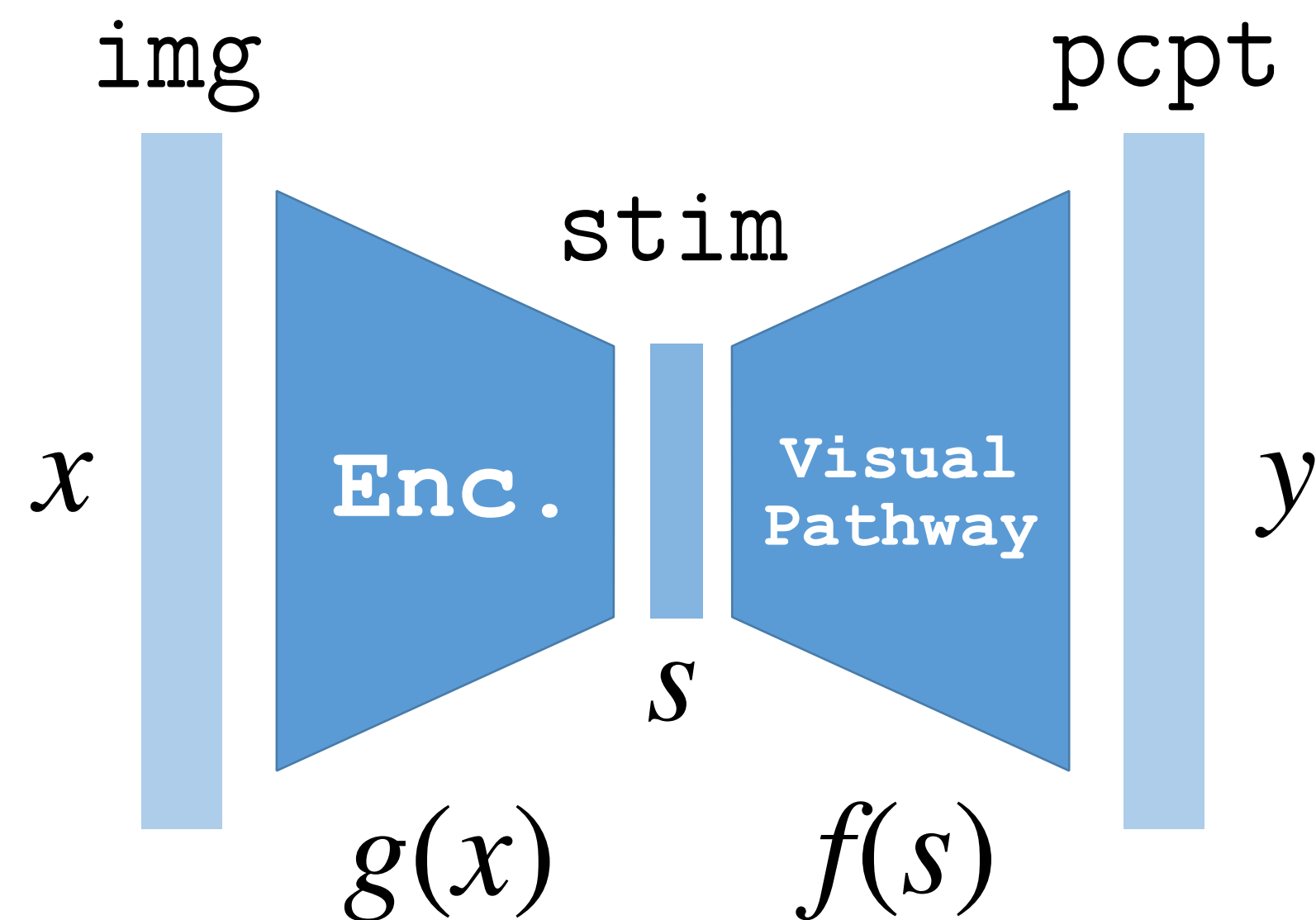
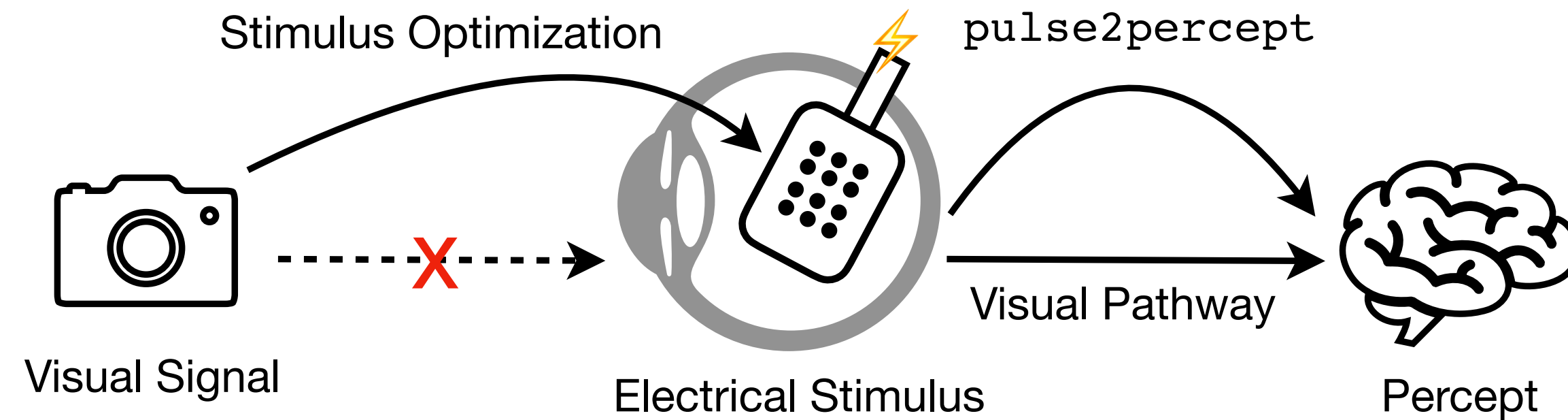
Invertible Neural Networks



Additional iterative reconstruction

[Wu et al., 2024b]

Encoding Summary



Reverse encoder:

$$g(x) := y \rightarrow s$$

Formalism

End-to-end encoder:

$$f(g(x)) \simeq x$$

Functionalism

Invertible encoder:

$$g := f^{-1}$$

Essentialism